

Hands-On Labs — CompTIA Certified A+ Training (Core 1 and Core 2)

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Register: <https://www.tertiarycourses.com.sg/wsqa-comptia-certified-a-training-core-1-and-core-2.html>

54 hands-on labs across the nine CompTIA A+ exam domains. **Each lab has its own folder** containing the full procedure (`README.md`) and a fill-in worksheet (`worksheet.md`).

Every lab runs on training hardware provided in class or in your web browser — see `tools.md` for the toolkit. There is nothing to install and nothing to license.

How to use these labs

1. Read the objective and the goal first, so you know what the lab is proving.
2. Work the steps in order — later steps depend on earlier ones.
3. Complete the **Test it** verification before moving on. It is what the Practical Performance assessment mirrors.
4. Fill in `worksheet.md` as you go and keep it as your evidence record.
5. Answer the review questions from memory — if you cannot, revisit the concept in the Learner Guide.

Core 1 (220-1101)

Topic 01 — Mobile Devices (13% of Core 1)

Laptop hardware · display components · accessories and ports · cellular and wireless · MDM and synchronisation

Lab	Title	Print	Exam objective
01	Laptop Teardown and Field-Replaceable Unit Identification	lab PDF · worksheet PDF	Identify and safely replace laptop field-replaceable units — battery, RAM, storage, keyboard and wireless card — with correct ESD precautions (Core 1 objective 1.1).
02	Mobile Display Technology and Digitizer Fault Diagnosis	lab PDF · worksheet PDF	Compare LCD, IPS, VA and OLED panel technologies and correctly distinguish a display panel fault from a backlight, inverter or digitizer fault (Core 1 objective 1.2).
03	Mobile Connectivity —	lab PDF ·	Configure and troubleshoot mobile connection

Lab	Title	Print	Exam objective
	Bluetooth, NFC, Hotspot and Cellular Configuration	worksheet PDF	methods including Bluetooth pairing, NFC, hotspot tethering and cellular data settings (Core 1 objectives 1.3 and 1.4).
04	MDM Policy Design and Mobile Synchronisation	lab PDF · worksheet PDF	Design a mobile device management policy and configure account synchronisation for corporate email, calendar and contacts, accounting for data caps (Core 1 objective 1.4).

Topic 02 — Networking (23% of Core 1)

Protocols and ports · network devices · wireless standards · SOHO networks · IP addressing · networking tools

Lab	Title	Print	Exam objective
05	Protocols and Ports — Building the A+ Port Reference	lab PDF · worksheet PDF	Identify the TCP/UDP ports, transport protocol and secure replacement for every protocol on the CompTIA A+ Core 1 objective 2.1 list.
06	IPv4 Subnetting and Address Planning with IP Calculator	lab PDF · worksheet PDF	Calculate subnet masks, network and broadcast addresses, usable host ranges and CIDR notation, and design an address plan for a SOHO network (Core 1 objective 2.5).
07	Network Devices and Traffic Behaviour Analysis	lab PDF · worksheet PDF	Distinguish hub, switch, router, firewall and access point by the layer they operate at and the traffic behaviour each produces (Core 1 objectives 2.2 and 2.3).
08	Wireless Standards, Channel Planning and Interference	lab PDF · worksheet PDF	Compare 802.11 standards and frequency bands, and design a channel plan that avoids co-channel and adjacent-channel interference (Core 1 objective 2.3).
09	SOHO Network Configuration — DHCP, DNS, NAT and Port Forwarding	lab PDF · worksheet PDF	Configure a SOHO router including DHCP scope, DNS, NAT, DHCP reservations and port forwarding, and verify each service from a client (Core 1 objective 2.5).
10	Network Cabling — Termination, Testing and Tool Selection	lab PDF · worksheet PDF	Terminate twisted-pair cable to T568B, test it end to end, and select the correct tool for each cabling task (Core 1 objectives 2.3 and 3.1).
11	Packet Capture Analysis —	lab PDF ·	Analyse a packet capture to identify protocols,

Lab	Title	Print	Exam objective
	Reading Real Network Traffic	worksheet PDF	conversations and anomalies, and use the evidence to support a network diagnosis (Core 1 objectives 2.1 and 5.7).

Topic 03 — Hardware (25% of Core 1)

Cables and connectors · RAM · storage and RAID · motherboards · CPUs and cooling · power supplies · printers

Lab	Title	Print	Exam objective
12	Cables and Connectors — Identification and Selection	lab PDF · worksheet PDF	Identify every cable and connector on the A+ objective list by sight and select the correct one for a stated requirement (Core 1 objectives 3.1 and 3.2).
13	RAM Identification, Installation and Channel Configuration	lab PDF · worksheet PDF	Identify RAM types and form factors, install modules correctly and configure dual-channel operation, verifying the result in firmware and the OS (Core 1 objective 3.3).
14	Storage Devices and RAID Level Selection	lab PDF · worksheet PDF	Compare HDD, SSD, NVMe and hybrid storage, and select the correct RAID level for a stated capacity, performance and redundancy requirement (Core 1 objectives 3.3 and 3.4).
15	Motherboard, BIOS/UEFI and CMOS Configuration	lab PDF · worksheet PDF	Identify motherboard form factors, expansion slots and headers, and configure BIOS/UEFI settings including boot order, virtualization support, Secure Boot and TPM (Core 1 objectives 3.4 and 3.5).
16	CPU Architecture, Sockets and Cooling Solutions	lab PDF · worksheet PDF	Compare CPU architectures, socket types and cooling solutions, and diagnose thermal problems from observed behaviour (Core 1 objectives 3.4 and 5.2).
17	Power Supply Sizing, Connectors and Safety	lab PDF · worksheet PDF	Select a power supply by wattage, form factor, efficiency and connector complement, and apply electrical safety rules when working with power (Core 1 objectives 3.5 and 4.5).
18	Printer Installation, Configuration and the Laser Imaging Process	lab PDF · worksheet PDF	Install and share a printer, configure its settings and security, and map each stage of the laser imaging process to the print defect its failure produces (Core 1 objectives 3.6 and 3.7).

Topic 04 — Virtualization and Cloud Computing (11% of Core 1)

Hypervisors · virtual machines · resource requirements · cloud models · shared resources and metered use

Lab	Title	Print	Exam objective
19	Hypervisor Types and Virtual Machine Resource Planning	lab PDF · worksheet PDF	Distinguish Type 1 from Type 2 hypervisors, plan virtual machine resource allocation and identify the firmware prerequisite for virtualization (Core 1 objective 4.2).
20	Building and Configuring a Linux Virtual Machine	lab PDF · worksheet PDF	Provision, configure and verify a Linux virtual machine, and perform the post-installation tasks a technician must complete on any new guest (Core 1 objective 4.2).
21	Cloud Service and Deployment Model Selection	lab PDF · worksheet PDF	Compare IaaS, PaaS and SaaS and the public, private, community and hybrid deployment models, and select the correct combination for a stated business requirement (Core 1 objective 4.1).

Topic 05 — Hardware and Network Troubleshooting (28% of Core 1)

The CompTIA troubleshooting methodology · POST and boot faults · storage and RAID · display faults · printer and network symptoms

Lab	Title	Print	Exam objective
22	Applying the CompTIA Six-Step Troubleshooting Methodology	lab PDF · worksheet PDF	Apply the CompTIA six-step troubleshooting methodology in order to a real fault, documenting each step (Core 1 objective 5.1).
23	POST, Boot and Power Fault Diagnosis	lab PDF · worksheet PDF	Diagnose no-power, no-POST and no-boot faults using beep codes, diagnostic LEDs and a systematic elimination sequence (Core 1 objective 5.2).
24	Storage and RAID Fault Diagnosis with S.M.A.R.T.	lab PDF · worksheet PDF	Diagnose storage faults using S.M.A.R.T. data, symptom analysis and RAID status, and choose the correct recovery action for each RAID level (Core 1 objective 5.3).
25	Display and Projector Fault Diagnosis	lab PDF · worksheet PDF	Diagnose display and projector faults by isolating source, cable, panel, backlight and settings as distinct failure points (Core 1 objective 5.4).
26	Mobile Device Hardware Fault Diagnosis	lab PDF · worksheet PDF	Diagnose mobile device hardware faults including battery, charging, screen, digitizer, port and overheating problems, applying the correct safety

Lab	Title	Print	Exam objective
			response (Core 1 objective 5.5).
27	Printer Fault Diagnosis from Print Defects	lab PDF · worksheet PDF	Diagnose printer faults by reading the physical defect on the page and mapping it to the failed component in the print process (Core 1 objective 5.6).
28	Network Fault Diagnosis — Command Line to Packet Level	lab PDF · worksheet PDF	Diagnose network faults by working the layers in order, from physical connectivity through addressing and DNS to application behaviour (Core 1 objective 5.7).

Core 2 (220-1102)

Topic 06 — Operating Systems (28% of Core 2)

Windows editions and installation · command line · Windows tools and MMC · file systems · macOS and Linux

Lab	Title	Print	Exam objective
29	Windows Editions, Installation Types and Partitioning	lab PDF · worksheet PDF	Select the correct Windows edition and installation type for a requirement, and plan MBR or GPT partitioning to match the firmware mode (Core 2 objectives 1.1 and 1.9).
30	Windows Command Line — Navigation, Files and Copy Operations	lab PDF · worksheet PDF	Use the Windows command line for navigation, file management and resilient copy operations, choosing correctly between copy, xcopy and robocopy (Core 2 objective 1.2).
31	Windows Networking and Repair Commands	lab PDF · worksheet PDF	Use Windows networking and repair commands to diagnose connectivity and repair system file corruption (Core 2 objective 1.2).
32	Windows Management Tools — Task Manager, MMC and Snap-ins	lab PDF · worksheet PDF	Use Task Manager, the Microsoft Management Console and its snap-ins to diagnose performance, inspect logs and manage devices, disks, users and tasks (Core 2 objectives 1.3 and 1.4).
33	File Systems, Permissions and Share Configuration	lab PDF · worksheet PDF	Compare NTFS, FAT32 and exFAT, and configure NTFS and share permissions correctly, applying the most-restrictive rule (Core 2 objectives 1.9 and 2.5).
34	Linux Command Line	lab PDF ·	Use the Linux commands named in Core 2 objective

Lab	Title	Print	Exam objective
	Essentials on Killercoda	worksheet PDF	1.11 for navigation, file management, searching, process control and package management (Core 2 objective 1.11).
35	Linux Permissions, Ownership and User Management	lab PDF · worksheet PDF	Apply the Linux permission model to users, groups and others using symbolic and numeric notation, and manage users and sudo rights (Core 2 objective 1.11).
36	macOS Tools and Cross-Platform File System Compatibility	lab PDF · worksheet PDF	Identify macOS features and utilities and resolve cross-platform file system compatibility between Windows, macOS and Linux (Core 2 objectives 1.10 and 1.9).

Topic 07 — Security (28% of Core 2)

Physical and logical security · authentication and MFA · wireless security · malware · social engineering · workstation and mobile hardening

Lab	Title	Print	Exam objective
37	Physical and Logical Security Control Design	lab PDF · worksheet PDF	Select and justify physical and logical security controls for a site, applying the principle of least privilege and defence in depth (Core 2 objectives 2.1 and 2.2).
38	Password Strength Analysis and Authentication Policy	lab PDF · worksheet PDF	Analyse password strength quantitatively and write an authentication policy grounded in entropy and crack-time evidence (Core 2 objectives 2.2 and 2.6).
39	Phishing Recognition and Social Engineering Defence	lab PDF · worksheet PDF	Identify phishing indicators and social engineering tactics, and design the user awareness training that defends against them (Core 2 objectives 2.4 and 2.7).
40	Malware Types, Symptoms and the Seven-Step Removal Procedure	lab PDF · worksheet PDF	Identify malware types by behaviour and apply the CompTIA seven-step malware removal procedure in order (Core 2 objectives 2.3 and 3.3).
41	Network Attack Recognition — Injection, XSS and Data Leakage	lab PDF · worksheet PDF	Recognise common application and network attacks including SQL injection, cross-site scripting, on-path attacks and denial of service, and state the defence for each (Core 2 objective 2.4).
42	Wireless Security and SOHO Router Hardening	lab PDF · worksheet PDF	Configure wireless encryption and harden a SOHO router against the attacks the default configuration invites (Core 2 objectives 2.9 and 2.10).

Lab	Title	Print	Exam objective
43	Windows Security Configuration and BitLocker Encryption	lab PDF · worksheet PDF	Configure Windows security settings including Defender, the firewall, UAC, account policy and BitLocker encryption (Core 2 objectives 2.5 and 2.6).
44	Data Destruction, Disposal and Regulated Data Handling	lab PDF · worksheet PDF	Select the correct data destruction method for a stated sensitivity and retention requirement, and handle regulated data according to its legal obligations (Core 2 objectives 2.8 and 4.6).

Topic 08 — Software Troubleshooting (23% of Core 2)

Windows OS symptoms · recovery tools · malware removal · browser problems · mobile OS and application faults

Lab	Title	Print	Exam objective
45	Windows OS Symptom Diagnosis and Recovery Tools	lab PDF · worksheet PDF	Diagnose common Windows symptoms and select the correct recovery tool, escalating from least to most destructive (Core 2 objectives 3.1 and 3.2).
46	Log Analysis with Regular Expressions	lab PDF · worksheet PDF	Filter and parse system logs with regular expressions to isolate the events that matter during troubleshooting (Core 2 objectives 3.1 and 4.8).
47	Malware Symptom Response and Browser Problem Resolution	lab PDF · worksheet PDF	Recognise malware and browser symptoms, apply the containment sequence and resolve browser-specific problems (Core 2 objectives 3.2 and 3.3).
48	Mobile OS and Application Troubleshooting	lab PDF · worksheet PDF	Diagnose and resolve mobile operating system and application faults including crashes, connectivity, battery and performance problems (Core 2 objectives 3.4 and 3.5).
49	System Restore, Backup Verification and Recovery Testing	lab PDF · worksheet PDF	Configure System Restore and Windows backup, then verify recovery by performing an actual restore (Core 2 objectives 3.1 and 4.3).

Topic 09 — Operational Procedures (21% of Core 2)

Documentation and ticketing · asset management · change management · backup and recovery · safety and environment · professionalism · scripting and remote access

Lab	Title	Print	Exam objective
50	Ticketing, Documentation and Knowledge Base	lab PDF · worksheet	Write support tickets and documentation to professional standard, and author a knowledge base

Lab	Title	Print	Exam objective
	Authoring	PDF	article from a resolved incident (Core 2 objective 4.1).
51	Change Management and Asset Management	lab PDF · worksheet PDF	Prepare a change request with risk analysis, rollback plan and approval path, and maintain an asset register through the procurement life cycle (Core 2 objectives 4.2 and 4.1).
52	Backup Strategy Design and the 3-2-1 Rule	lab PDF · worksheet PDF	Design a backup strategy specifying types, rotation, retention and off-site copy, and justify it against a stated recovery objective (Core 2 objective 4.3).
53	Safety Procedures, ESD Control and Environmental Compliance	lab PDF · worksheet PDF	Apply ESD, electrical and personal safety procedures, and comply with environmental requirements for disposal and equipment protection (Core 2 objectives 4.4 and 4.5).
54	Professional Communication, Scripting and Remote Access	lab PDF · worksheet PDF	Apply professional communication standards including handling difficult customers, and select appropriate scripting and remote access methods for support tasks (Core 2 objectives 4.7, 4.8 and 4.9).

Lab folder structure

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labs/

lab-01-laptop-teardown-and-field-replaceable-unit-identification/

    README.md      the full lab procedure

    worksheet.md    fill-in evidence record

lab-02-mobile-display-technology-and-digitizer-fault-diagnosis/

    README.md      the full lab procedure

    worksheet.md    fill-in evidence record

lab-03-mobile-connectivity-bluetooth-nfc-hotspot-and-cellular-configuration/

    README.md      the full lab procedure

    worksheet.md    fill-in evidence record

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(54 lab folders in total)

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<code>README.md</code>	<code>this index</code>
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<code>tools.md</code>	<code>the lab toolkit reference</code>
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